HSA: from CPU to Metal

One model, four implementation stages. Decode latency is measured over the entire model, not an isolated HSA kernel.

CPU HSA

8b9b27a / CPU backend

CPU scoring, Top-K and exact sparse attention.

2K	**422.25 ms**
4K	**596.40 ms**
8K	**799.08 ms**
16K	**1062.37 ms**

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Initial Metal

8b9b27a / Metal backend

Move HSA kernels to GPU; retain per-layer staging / waits.

2K	**51.48 ms**
4K	**75.67 ms**
8K	**102.82 ms**
16K	**159.68 ms**

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KV / graph

a708c7c

Direct F16 KV + cached chunks; threadgroup sort + GGML encoder.

2K	**25.65 ms**
4K	**32.62 ms**
8K	**32.52 ms**
16K	**50.16 ms**

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Final Metal

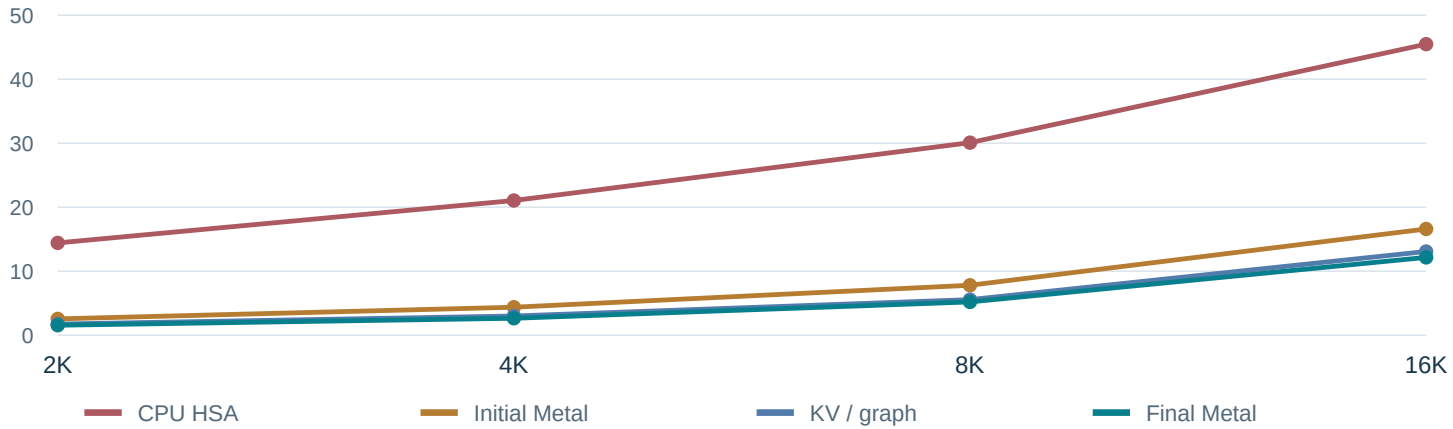
dbf4743

Chunk sorting, 64-thread groups, four-V loads and SIMD shuffle.

2K	**20.42 ms**
4K	**21.54 ms**
8K	**21.32 ms**
16K	**21.40 ms**

WHOLE PIPELINE LATENCY

Prefill + Stage 1 + initialization + 32 decode steps | seconds, lower is better



CPU -> FINAL

Pipeline time (seconds)

2K	**14.42 -> 1.56**	**9.24x**
4K	**21.05 -> 2.65**	**7.95x**
8K	**30.08 -> 5.20**	**5.78x**
16K	**45.47 -> 12.16**	**3.74x**

FINAL KERNEL REFINEMENTS

Incremental decode-time reduction | prior 3-round A/B | 8K / 16K

Sort chunks, expand IDs

8K: 32.47 -> 26.70 ms	**-17.77%**
16K: 50.29 -> 26.79 ms	**-46.73%**

64-thread weighted sum

8K: 26.61 -> 26.12 ms	**-1.83%**
16K: 26.84 -> 26.26 ms	**-2.17%**

Preload four V values

8K: 26.17 -> 21.48 ms	**-17.93%**
16K: 26.25 -> 21.76 ms	**-17.11%**

SIMD shuffle sorting

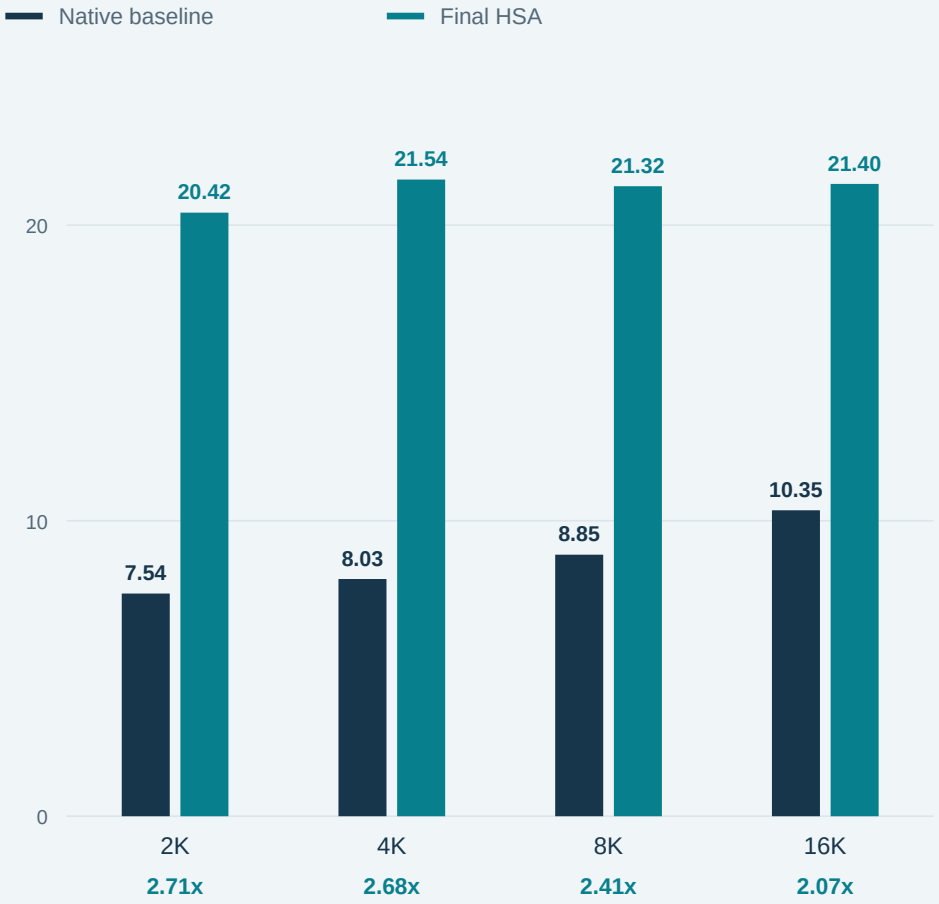
8K: 21.58 -> 21.32 ms	**-1.20%**
16K: 21.72 -> 21.35 ms	**-1.69%**

Latency: one run per stage / length; identical forced tokens; M2 Ultra, Llama-3.2-3B Q8_0, F16 KV, GPU layers=99, budget=1024, threads=16.
Model stays on GPU for CPU HSA. Pipeline excludes loading / tokenization / sampling. Each kernel card is a separate A/B. Sources: benchmark JSON + HSA_OPTIMIZATION_LOG.md.

Final HSA vs native baseline

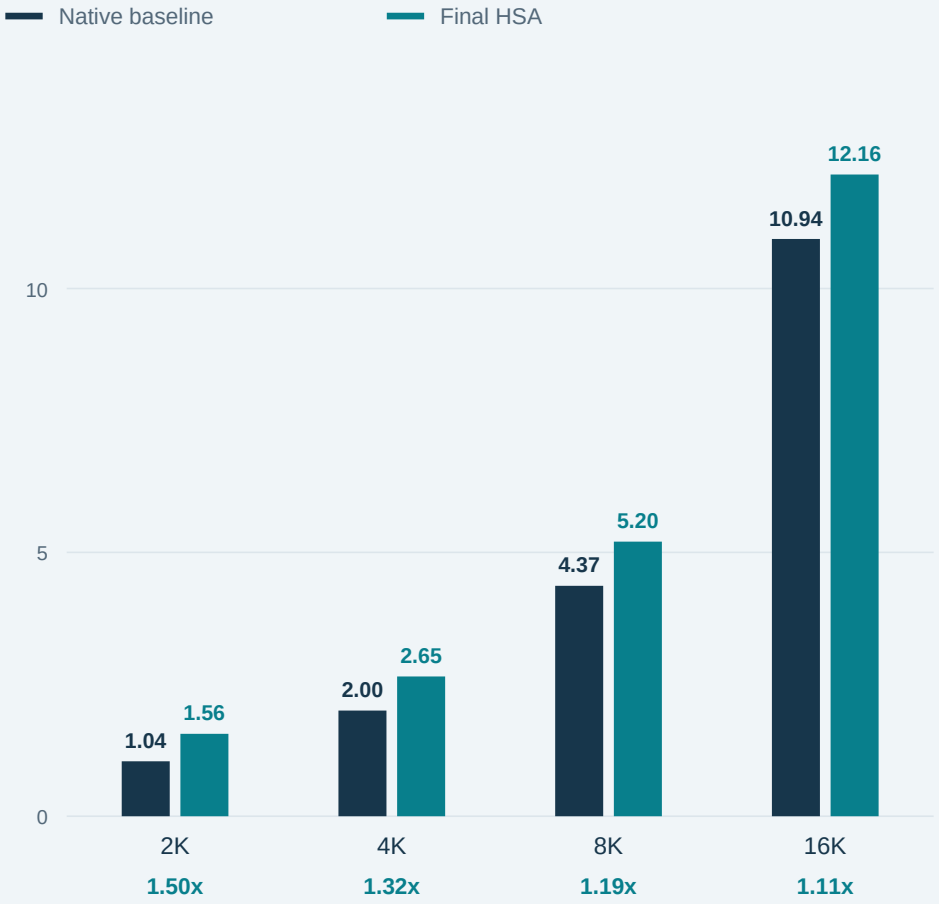
Final Metal dbf4743 vs native full attention | 2K / 4K / 8K / 16K | Lower latency is better.

Decode latency (ms / forward)



Ratio below each group = final / baseline latency (>1 means slower).

Pipeline latency (s)



Ratio below each group = final / baseline latency (>1 means slower).

Result: GPU migration accelerates HSA, but final HSA still has higher latency than native full attention.

Latency: one run per configuration, identical forced tokens, 32 decode forwards. Pipeline = prefill + Stage 1 + initialization + decode.
Pipeline excludes model loading, tokenization and sampling. Sources: docs/hsa_pipeline_benchmark.json; HSA_OPTIMIZATION_LOG.md.